

Nil Blanco

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EDUCATION

Technical University of Munich (TUM)

Erasmus Exchange Program (BSc Physics)

Munich, Germany

Oct 2024 – Sep 2025

- Selected for a competitive 11-month academic exchange at one of Europe's top-tier technical universities.
- Adapted to a high-performance international environment, completing advanced coursework in physical sciences.

Universitat de Barcelona (UB)

Bachelor of Science in Physics

Barcelona, Spain

Sep 2021 – Jan 2026

- Developed strong analytical and problem-solving skills through rigorous mathematical modeling and theoretical physics.

TECHNICAL & RESEARCH PROJECTS

Numerical Simulation of Gravitational Waves (LISA Mission)

Data Analysis & Mathematical Modeling

Academic Research

Nov 2025 – Present

- Developed numerical models to simulate complex physical signals and timing residuals related to the Laser Interferometer Space Antenna (LISA).
- Applied Schwarzschild-de Sitter metrics to solve advanced non-linear equations, demonstrating capacity for high-level abstract mathematical reasoning.
- Manuscript in preparation:** Currently co-authoring a comprehensive research paper based on these findings for publication in a peer-reviewed scientific journal, in collaboration with researchers from ICCUB.

Algorithmic Trading Bot Development

Quantitative Trading & Backtesting

Personal Project

May 2026 – Present

- Programmed a custom algorithmic trading bot in Python to simulate and evaluate automated market strategies.
- Developed and backtested Long/Short and Parabolic SAR trading models utilizing extensive historical market data.
- Applied data analysis and statistical validation to optimize technical indicators and risk management parameters.

PROFESSIONAL EXPERIENCE

Generalitat de Catalunya

Mathematics Educator

Catalonia, Spain

Mar 2026 – Present

- Translate complex mathematical concepts into clear, accessible frameworks for diverse audiences.
- Design, structure, and format quantitative assessments and technical documentation using LaTeX.
- Manage classroom dynamics and optimize learning processes, demonstrating strong communication and leadership skills.

SKILLS

- Technical & Programming:** Advanced LaTeX, Mathematical Modeling, Data Analysis, Numerical Simulation, Python.
- Languages:** English (Professional Working Proficiency - *C1/C2 certification in progress*), Spanish (Native), Catalan (Native).
- Core Competencies:** Quantitative Analysis, Complex Problem Solving, Cross-cultural Adaptability, Self-directed Learning.